

Unit 2 Guided Notes — Exponents & Scientific Notation

Same base? Move the exponents. Big or tiny number? Pack it into a $\times 10^n$.

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

This packet weaves three parts together: **Guided Notes** with worked examples, **Practice** built into each section to try as you learn, and a **Cumulative Review** with a full **Answer Key** at the end. Fill in the blanks and show your work on your printed copy.

Topics Covered: 2.1 Exponent Rules — Product, Quotient, Power, Zero, Negative (NC.8.EE.1) · 2.2 Scientific Notation — Format & Conversions (NC.8.EE.3) · 2.2 Multiplying & Dividing Scientific Notation (NC.8.EE.4) · 2.2 Word Problems with Scientific Notation (NC.8.EE.4)

⚠ **Parentheses come first!** A power attaches only to what it touches. $-3^2 = -9$, but $(-3)^2 = 9$. And a power outside parentheses hits EVERYTHING inside: $(3x^2)^3 = 27x^6$ (the 3 is cubed too). Keep an eye on parentheses all unit long.

1 Expanding & Condensing Expressions

➤ **KEY IDEA:** A power is repeated multiplication. Expand = write out every factor. Condense = write repeated factors as a single power.

See it in Action

Expand · $x^4 = x \cdot x \cdot x \cdot x$

$$x^4 \rightarrow x \cdot x \cdot x \cdot x$$

The exponent 4 means multiply x four times $\rightarrow x \cdot x \cdot x \cdot x$.

Condense · $x \cdot x \cdot x \cdot x \cdot x = x^5$

$$x \cdot x \cdot x \cdot x \cdot x \rightarrow x^5$$

Five repeated factors $\rightarrow$ write them as one power, x^5 .

Compound & Rational Bases

Compound base: $(2x)^3 = (2x)(2x)(2x) = 8x^3$ · $(xy)^2 = xy \cdot xy = x^2y^2$

Rational: $(\frac{x}{y})^3 = (\frac{x}{y})(\frac{x}{y})(\frac{x}{y}) = \frac{x^3}{y^3}$ · condense $\frac{x \cdot x \cdot x}{y \cdot y} = \frac{x^3}{y^2}$

Worked Example — Condense $4 \cdot a \cdot a \cdot a \cdot b \cdot b$

Step 1 — Coefficient: the 4 stays out front

Step 2 — Count the a's: $a \cdot a \cdot a = a^3$

Step 3 — Count the b's: $b \cdot b = b^2$

✓ **Combine:** $4a^3b^2$

PRACTICE 1 Expand & Condense

Expand = write out the factors · Condense = write as a power. Keep quotients as reduced fractions. **Show your work in the box.**

1Expand: x^6 **2**Expand: $4a^3$ **3**Expand: $2y^5$ **4**Condense: $x \cdot x \cdot x \cdot x \cdot x$ **5**Condense: $3 \cdot b \cdot b \cdot b \cdot b$ **6**Condense: $m \cdot m \cdot m \cdot m \cdot m \cdot m \cdot m$ **7**Expand: $(2x)^3$ **8**Condense: $(5a)(5a)$ **9**Condense: $(xy)(xy)(xy)(xy)$ **10**Condense: $\frac{x \cdot x \cdot x}{y \cdot y}$ **11**Condense: $\frac{2 \cdot a \cdot a}{b \cdot b \cdot b}$ **12**Condense: $\frac{m \cdot m}{n \cdot n \cdot n \cdot n}$ **2 Combining Like Terms**

➤ **KEY IDEA:** Like terms have the SAME variables with the SAME exponents. Combine by adding/subtracting the coefficients; keep the variable part the same.

See it in Action

Combine · $3x^2 + 5x^2 = 8x^2$

$$\boxed{x^2} \boxed{x^2} \boxed{x^2} + \boxed{x^2} \boxed{x^2} \boxed{x^2} \boxed{x^2} \boxed{x^2} = \boxed{8x^2}$$

Like terms → add the coefficients: $3 + 5 = 8$, so $8x^2$.

Unlike · $4x^2 + 3x$ cannot combine

$$\boxed{x^2} \boxed{x^2} \boxed{x^2} \boxed{x^2} + \boxed{x} \boxed{x} \boxed{x} \rightarrow \boxed{4x^2 + 3x}$$

x^2 and x are NOT like terms → the expression stays $4x^2 + 3x$.

Watch Out

Different variables or exponents = NOT like terms: $5x + 3y$ stays $5x + 3y$ · $2x^2 + 4x$ stays $2x^2 + 4x$

Longer expressions — group first: $4x^2 + 2x + 3x^2 - x = (4x^2 + 3x^2) + (2x - x) = 7x^2 + x$

Worked Example — Simplify $5a^2 + 3a - 2a^2 + 7a$

Step 1 — Group like terms: a^2 terms and a terms

Step 2 — Combine a^2 : $5a^2 - 2a^2 = 3a^2$

Step 3 — Combine a : $3a + 7a = 10a$

✓ **Combine:** $3a^2 + 10a$

PRACTICE 2 Combining Like Terms

Combine like terms only. If terms are not alike, leave the expression as is.

1

Simplify $3x + 5x$

2

Simplify $9y - 4y$

3

Simplify $2a^2 + 6a^2$

4

Simplify $7mn - 3mn$

5

Simplify $4x^2y + 5x^2y$

6

Simplify $10p - 3p$

7 Simplify $6c^3 + 5c^3$

8 Simplify $5x + 3y$

9 Simplify $2x^2 + 4x$

10 Simplify $6a^2 + 3a - 2a^2$

11 Simplify $3m^2 + 5m - m^2 + 2m$

12 Simplify $8a - 3b + 2a + b$

3 Product & Quotient Rules

➤ **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Fill the examples and key phrases on your printed copy.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	_____	_____	_____
Quotient Rule	$x^a \div x^b = x^{a-b}$	_____	_____	_____

PRODUCT RULE: COUNT THE FACTORS

$$\begin{array}{c} \boxed{x} \boxed{x} \boxed{x} \\ x^3 \text{ (3 factors)} \end{array} \times \begin{array}{c} \boxed{x} \boxed{x} \\ x^2 \text{ (2)} \end{array} = \begin{array}{c} \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} \\ x^5 \text{ (5 factors)} \end{array}$$

$$x^3 \times x^2 = x^{3+2} = x^5 \rightarrow \text{ADD the exponents}$$

See WHY

Product Rule · $x^3 \times x^2 = x^5$

See why → $\boxed{x} \boxed{x} \boxed{x} \times \boxed{x} \boxed{x} = \boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} = x^5$

Count the factors: $3 + 2 = 5$. Same base → ADD the exponents.

Quotient Rule · $x^5 \div x^2 = x^3$

See why → $\boxed{x} \boxed{x} \boxed{x} \boxed{x} \boxed{x} \div \boxed{x} \boxed{x} \rightarrow \text{cancel 2 pairs} \rightarrow \boxed{x} \boxed{x} \boxed{x} = x^3$

Cancel matching factors: $5 - 2 = 3$. Same base → SUBTRACT the exponents.

Quick Check

MC Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?

☐ Product Rule $\rightarrow x^8$ ☐ Power Rule $\rightarrow x^{15}$ ☐ Quotient Rule $\rightarrow x^2$

Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 — Multiply coefficients: $4 \times 3 = 12$

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

PRACTICE 3 Product & Quotient

Rules: Multiply $\rightarrow$ ADD exponents · Divide $\rightarrow$ SUBTRACT. Write each quotient as a reduced fraction. **Show your work in the box.**

1

Simplify $x^4 \cdot x^7$

2

Simplify $(3x^2)(5x^6)$

3

Simplify $(2a^3b^2)(4a^2b^5)$

4

Simplify $(-3x^4y)(6xy^3)$

5

Simplify $(5m^2n)(2mn^3)(3n^2)$

6

Simplify $\frac{x^9}{x^4}$

7

Simplify $\frac{15a^7}{3a^2}$

8

Simplify $\frac{24x^5y^6}{6x^2y^2}$

9

Simplify $\frac{20m^8n^3}{5m^3n}$

10

Simplify $\frac{-18p^6q^5}{3p^2q^2}$

11

Simplify $\frac{(2x^3)(6x^4)}{4x^2}$

12

Simplify $\frac{(3a^2b)(4a^3b^2)}{6ab}$

4 Power Rules

A power outside parentheses applies to every factor inside. Fill in, then check.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Power Rule	$(x^a)^b = x^{a \times b}$	_____	_____	_____
Power of a Product	$(xy)^n = x^n y^n$	_____	_____	_____
Power of a Quotient	$(\frac{x}{y})^n = \frac{x^n}{y^n}$	_____	_____	_____

⚠ PARENTHESES MATTER!

$$-3^2 = \underline{\hspace{2cm}} \quad \text{vs.} \quad (-3)^2 = \underline{\hspace{2cm}}$$

Without (), ONLY the base gets the exponent — not the negative sign!

See WHY

Power Rule • $(x^2)^3 = x^6$

See why → $x^2 \cdot x^2 \cdot x^2 = x \cdot x \cdot x \cdot x \cdot x \cdot x = x^6$

Three copies of x^2 : $2 \times 3 = 6$. Power to a power → MULTIPLY.

ERROR ANALYSIS Spot the Error — Simplify $(3x^2)^3$. A student wrote $3x^6$.

The coefficient 3 must ALSO be cubed: $3^3 = 27$. Correct: $27x^6$. The power rule applies to EVERYTHING inside the parentheses.

Worked Example — Simplify $(2a^3b)^4$

Step 1 — Power the coefficient: $2^4 = 16$

Step 2 — Multiply the a exponent: $a^{3 \times 4} = a^{12}$

Step 3 — Multiply the b exponent: $b^{1 \times 4} = b^4$

✓ **Step 4 — Combine:** $16a^{12}b^4$

PRACTICE 4 Power Rules

Rule: Power to a power → MULTIPLY; apply the power to EVERY factor, top and bottom.

1

Simplify $(x^4)^5$

2

Simplify $(2x^3)^4$

3

Simplify $(-3a^2)^3$

4

Simplify $(x^2y^5)^3$

5

Simplify $(4m^3n^2)^2$

6

Simplify $(2x^3y^4)^5$

7

Simplify $(\frac{x^2}{y^3})^4$

8

Simplify $(\frac{2a}{b^2})^3$

9

Simplify $(\frac{3x^4}{2y})^2$

10

Simplify $((x^3)^2)^4$

11

Simplify $(2x^2y)^3(3xy^2)^2$

12

Simplify $(-2a^3b^2)^4$

5 Zero & Negative Exponents

Anything to the 0 power is 1; a negative exponent flips to a fraction.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Zero Exponent	$x^0 = 1$	_____	_____	_____
Negative Exponent	$x^{-n} = \frac{1}{x^n}$	_____	_____	_____

See WHY

Zero Exponent · $x^0 = 1$

See why → $x \cdot x \cdot x \div x \cdot x \cdot x = 1$, but by the quotient rule = $x^{3-3} = x^0$. So $x^0 = 1$

Anything (except 0) divided by itself is 1 → $x^0 = 1$.

Negative Exponent · $x^{-3} = \frac{1}{x^3}$

See why → $x \cdot x \div x \cdot x \cdot x \cdot x = x^{2-5} = x^{-3}$; cancelling leaves $\frac{1}{x^3}$

A negative exponent means 'flip it' → $x^{-n} = \frac{1}{x^n}$.

Match the Rule to its Key Phrase

MATCH Draw a line from each rule to its key phrase.

Rule

Product Rule

Quotient Rule

Power Rule

Negative Exponent

Key Phrase

Power to power? MULTIPLY

Same base? ADD exponents

Flip to a fraction

Same base? SUBTRACT exponents

Negative Exponents Across the Fraction Bar

A negative exponent moves its factor to the OTHER side of the bar and turns positive.

Move across the bar: $x^{-3} = \frac{1}{x^3} \cdot \frac{1}{x^{-3}} = x^3 \cdot \frac{x^{-2}}{y^{-4}} = \frac{y^4}{x^2}$

In parentheses — flip to the reciprocal: $(\frac{a}{b})^{-n} = (\frac{b}{a})^n$. $(\frac{2}{3})^{-2} = (\frac{3}{2})^2 = \frac{9}{4} \cdot (\frac{x^2}{y})^{-3} = (\frac{y}{x^2})^3 = \frac{y^3}{x^6}$

PRACTICE 5 Zero & Negative Exponents

Write every answer with POSITIVE exponents only, as a reduced fraction.

1

Simplify $(7x^4y^2)^0$

2

Simplify $(3a^2b)^0$

3

Evaluate 4^{-3}

4

Positive exponents: $2x^{-5}$

5Simplify $\frac{x^{-3}}{y^{-2}}$ **6**Simplify $\frac{a^{-2}b^3}{c^{-1}}$ **7**Simplify $\frac{12m^{-2}n^4}{3m^4n^{-1}}$ **8**Simplify $\frac{8x^{-3}y^2}{2xy^{-1}}$ **9**Simplify $(\frac{x}{y})^{-2}$ **10**Evaluate $(\frac{2}{3})^{-3}$ **11**Simplify $(\frac{a^2}{b})^{-3}$ **12**Simplify $(\frac{3x}{y^2})^{-2}$

6 Scientific Notation

$$a \times 10^n \quad \text{where} \quad 1 \leq a < 10$$

4.5×10^7 | coefficient · base (always 10) · exponent

SCIENTIFIC NOTATION: MOVE THE DECIMAL

$$45,000 \rightarrow 4.5 \times 10^4$$

big number → move LEFT 4 places → exponent +4

$$0.0067 \rightarrow 6.7 \times 10^{-3}$$

small number → move RIGHT 3 places → exponent -3

Type	Direction	Exponent	Example
Large number	Move decimal LEFT	_____	$45,000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT	_____	$0.0067 = 6.7 \times 10^{-3}$

PRACTICE B Scientific Notation Conversions

Remember: $1 \leq a < 10$. Positive exp = BIG · Negative exp = SMALL.

1

45,000,000 → sci. notation

2

0.00032 → sci. notation

3

6.02×10^8 → standard

4

9.1×10^{-5} → standard

5

780,000 → sci. notation

6

0.0000067 → sci. notation

7

3.45×10^4 → standard

8

1.2×10^{-3} → standard

9

5,280,000,000 → sci. notation

10

 $2.5 \times 10^{-2} \rightarrow$ standard

11

0.000000904 → sci. notation

12

 $7.15 \times 10^6 \rightarrow$ standard

13

63,400,000,000 → sci. notation

14

 $8.8 \times 10^{-6} \rightarrow$ standard

Multiplying Scientific Notation — 3-Step Process

1. _____ the coefficients (the a-values)
2. _____ the exponents
3. Adjust if coefficient is NOT between _____ and _____

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients: $3 \times 2 = 6$

ADD exponents: $4 + 5 = 9$

✓ **Answer:** 6×10^9

Dividing Scientific Notation — 3-Step Process

1. _____ the coefficients
2. _____ the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

Divide coefficients: $8 \div 4 = 2$

SUBTRACT exponents: $6 - 2 = 4$

✓ **Answer:** 2×10^4

⚠ ADJUSTMENT RULES:

- If coefficient ≥ 10 : Move decimal LEFT, ADD 1 to exponent
- If coefficient < 1 : Move decimal RIGHT, SUBTRACT 1 from exponent

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

Multiply: $5 \times 4 = 20$

Add exponents: $3 + 2 = 5 \rightarrow$ result: 20×10^5

Adjust ($20 \geq 10$): $20 = 2.0 \times 10^1$, so ADD 1 to exponent

✓ **Final answer:** 2.0×10^6

ERROR ANALYSIS **Spot the Error — Convert 0.0045 to scientific notation.** A student wrote 4.5×10^3 .

Small number $\rightarrow$ NEGATIVE exponent. Correct: 4.5×10^{-3} . Numbers less than 1 always have negative exponents in scientific notation.

PRACTICE C Operations with Scientific Notation

Multiply/divide coefficients · add/subtract exponents · adjust so $1 \leq a < 10$.

1

$$(3 \times 10^4)(2 \times 10^5)$$

2

$$(8 \times 10^7) \div (4 \times 10^3)$$

3

$$(5 \times 10^3)(7 \times 10^6)$$

4

$$(9.6 \times 10^8) \div (3.2 \times 10^5)$$

5

$$(6 \times 10^5)(5 \times 10^4)$$

6

$$(1.5 \times 10^{-3})(4 \times 10^7)$$

7

$$(7.2 \times 10^9) \div (1.8 \times 10^{-2})$$

8

$$(2.5 \times 10^6)(8 \times 10^{-3})$$

Word Problems with Scientific Notation

Strategy: Identify what operation to use. Write your scientific-notation setup. Adjust if needed.

PRACTICE D Word Problems

Identify the operation, write your setup, then adjust so $1 \leq a < 10$.

1. The Sun is about 9.3×10^7 miles from Earth. Light travels 1.86×10^5 miles per second. How many seconds for light to reach Earth?

Answer: _____

2. A bacteria cell is about 2×10^{-6} m long; a virus about 5×10^{-8} m. How many times longer is the bacteria?

Answer: _____

3. The US debt is about 3.4×10^{13} dollars; population about 3.4×10^8 . What is the debt per person?

Answer: _____

4. A grain of sand has mass 6.5×10^{-4} g. A beach has 8×10^{18} grains. Total mass in grams?

Answer: _____

5. A body has 3.7×10^{13} cells, each 1×10^{-5} m wide. Lined up, how long in meters?

Answer: _____

6. A computer does 4.8×10^9 calculations per second. A task needs 1.44×10^{12} . How many seconds?

Answer: _____

7. A hard drive holds 2×10^{12} bytes. Each photo is 4×10^6 bytes. How many photos fit?

Answer: _____

8. Light travels 3×10^8 m/s. How far in 6×10^3 seconds?

Answer: _____

★ Cumulative Practice — Unit 2 Review ★

Mixed practice over everything above. Show all work; circle your final answer.

REVIEW · A Exponent Rules

1

Simplify $x^4 \cdot x^6$

2

Simplify $(3x^2)(4x^5)$

3

Simplify $(2a^3b^2)(5ab^4)$

4

Simplify $\frac{24x^7}{6x^3}$

5

Simplify $(m^3)^5$

6

Simplify $(2x^2y^3)^4$

7

Simplify $\frac{18a^5b^6}{3a^2b^2}$

8

Simplify $(\frac{x^4}{y^2})^3$

REVIEW · B Scientific Notation Conversions

1

7,200,000 → sci. notation

2

0.00045 → sci. notation

3

3.05×10^7 → standard

4

6.1×10^{-5} → standard

5
91,000 → sci. notation

6
0.0000008 → sci. notation

7
 $4.44 \times 10^6 \rightarrow$ standard

8
 $2.7 \times 10^{-3} \rightarrow$ standard

REVIEW · C **Operations**

1
 $(3 \times 10^4)(3 \times 10^5)$

2
 $(8 \times 10^9) \div (2 \times 10^3)$

3
 $(5 \times 10^6)(4 \times 10^3)$

4
 $(9.6 \times 10^8) \div (1.6 \times 10^2)$

5
 $(7 \times 10^5)(8 \times 10^2)$

6
 $(1.2 \times 10^{-3})(4 \times 10^5)$

REVIEW · D **Word Problems**

1. A red blood cell is about 8×10^{-6} m wide; a platelet about 2×10^{-6} m. How many times wider is the red blood cell?

Answer: _____

2. Earth is about 1.5×10^8 km from the Sun. A spacecraft travels 3×10^4 km/h. How many hours for the trip?

Answer: _____

3. A factory makes 5×10^3 widgets per day. How many in 2×10^2 days?

Answer: _____

4. A virus is about 3×10^{-7} m and a cell about 6×10^{-5} m. How many times larger is the cell?

Answer: _____

REVIEW · E Negative Exponents & Rational Expressions

Write every answer with positive exponents only, as a reduced fraction.

1
Evaluate 5^{-2}

2
Positive exponents: $3x^{-4}$

3
Simplify $\frac{x^{-5}}{y^{-3}}$

4
Simplify $(\frac{a}{b})^{-3}$

5
Simplify $\frac{10a^{-2}b^5}{2a^3b^{-1}}$

6
Simplify $(\frac{2x^2}{y})^{-2}$

REVIEW · F Expand, Condense & Like Terms

1

Condense: $x \cdot x \cdot x \cdot x \cdot x$

2

Expand: $(3a)^2$

3

Condense: $\frac{m \cdot m \cdot m}{n \cdot n}$

4

Simplify $4x + 7x$

5

Simplify $8a^2 - 3a^2$

6

Simplify $6mn + 2m - mn$